Inverse PDE Optimization using Genetic Algorithm and Particle Swarm Optimization(pso)

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1 Introduction

Inverse problems are used to find unknown parameters from observed data.

In this project, I study a simple one-dimensional advection-diffusion problem. The goal is to find the advection velocity v and the diffusion coefficient D from noisy observations.

The main equation used in the model is:

$$u_t + vu_x = Du_{xx}$$

where u is the transported quantity, v is the advection velocity, and D is the diffusion coefficient.

I use a numerical PDE solver and two optimization methods, Genetic Algorithm (GA) and Particle Swarm Optimization (PSO), to estimate the unknown parameters.

The observations are generated from known parameters and then 1% Gaussian noise is added.

2 Methodology

2.1 PDE Solver

I solve the one-dimensional advection-diffusion equation using the Finite Difference Method (FDM).

The advection term uses an upwind difference:

$$advection = -v(u[i] - u[i-1])/dx$$

The diffusion term uses a central difference:

$$diffusion = D(u[i+1] - 2u[i] + u[i-1])/dx^2$$

The solution is updated using:

$$u\_new[i] = u[i] + dt(advection + diffusion)$$

The initial condition is a Gaussian pulse:

$$u = exp(-100(x - 0.3)^2)$$

The grid uses 80 points, with:

$$dt = 0.0005$$

and:

$$Nt = 150$$

2.2 Inverse Problem

The goal is to find v and D that give a simulated solution close to the observed solution.

The loss function used in the code is:

$$J(v, D) = ||u\_sim(v, D) - u\_obs||$$

The optimization tries to make this value as small as possible.

2.3 Genetic Algorithm

I use a simple Genetic Algorithm (GA).

The initial population contains random values for v and D.

The algorithm selects the best solutions and creates new solutions using:

- selection
- crossover
- mutation

The values are also kept inside the allowed parameter bounds.

2.4 Particle Swarm Optimization

I also use Particle Swarm Optimization (PSO).

Each particle has a position and a velocity. The position contains the values of v and D.

The particle updates its velocity using its own best position and the global best position.

The main parameters used are:

$$w = 0.7, \quad c1 = 1.5, \quad c2 = 1.5$$

PSO is used to search for parameter values with a small loss.

3 Results

The model was tested using synthetic data. The true parameters were:

$$v_{true} = 0.8$$

$$D_{true} = 0.05$$

The observations were generated using these values and 1% Gaussian noise was added.

3.1 Genetic Algorithm Results

After 3 runs, the mean estimated values were:

$$v_{GA} = 0.7957$$

$$D_{GA} = 0.04993$$

The mean errors were:

$$Error_v = 0.01499$$

$$Error_D = 0.000261$$

3.2 PSO Results

After 3 runs, the mean estimated values were:

$$v_{PSO} = 0.7977$$

$$D_{PSO} = 0.04963$$

The mean errors were:

$$Error_v = 0.01266$$

$$Error_D = 0.000365$$

3.3 Comparison

The results show that both methods can recover values close to the true parameters.

PSO gives a slightly smaller error for the velocity v.

GA gives a slightly smaller error for the diffusion coefficient D.

For these 3 runs, the difference between the two methods is small.

3.4 Observation

Both GA and PSO were able to estimate the unknown parameters from noisy observations.

The results are close to the true values:

$$v = 0.8$$

$$D = 0.05$$

The results can change between runs because both optimization methods use random values.

4 Conclusion

In this project, I used a numerical PDE solver to estimate unknown parameters in a one-dimensional advection-diffusion problem.

I used two optimization methods: Genetic Algorithm (GA) and Particle Swarm Optimization (PSO).

The true parameters were:

$$v = 0.8$$

$$D = 0.05$$

After 3 runs, both methods gave values close to the true parameters.

PSO gave a slightly better result for the velocity, while GA gave a slightly better result for the diffusion coefficient in these runs.

This project shows that numerical simulation and simple optimization methods can be used to solve an inverse PDE problem.

Future work can include more runs, more noise levels, higher-dimensional PDE models, and other optimization methods.